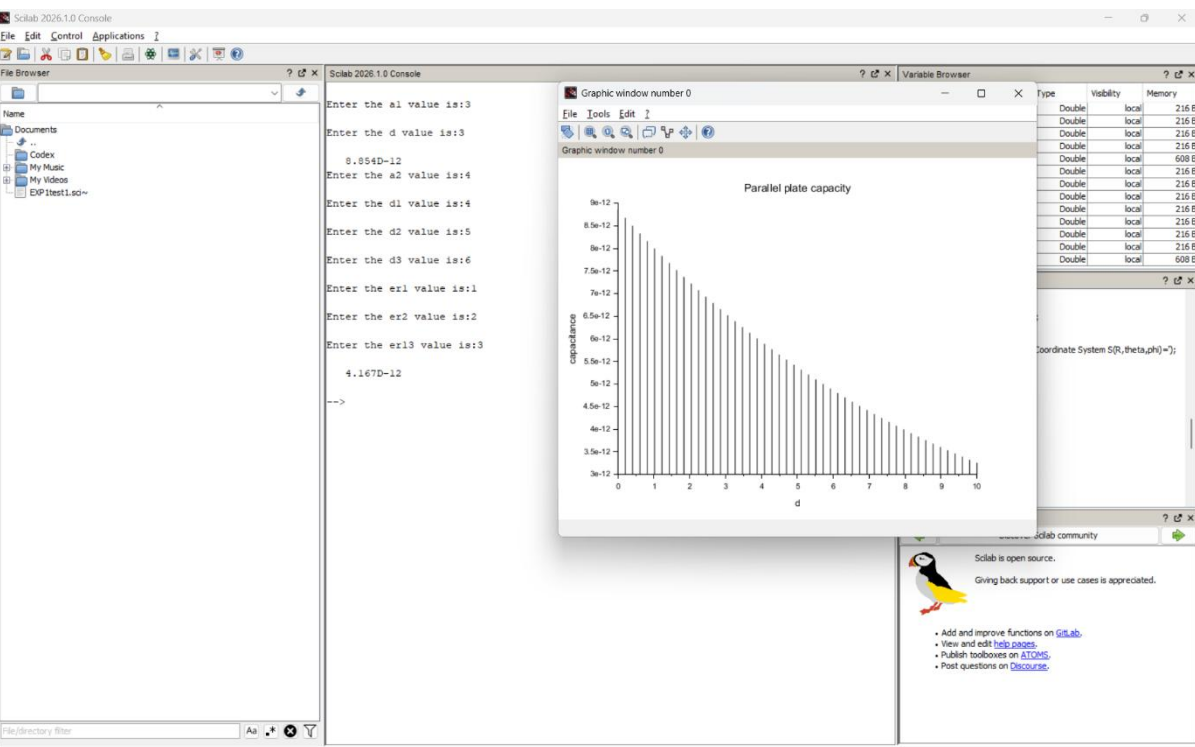


The screenshot shows a MATLAB Scinotes editor window titled "Untitled 1 - Scinotes". The script defines a function to calculate the capacitance of a parallel plate capacitor with three dielectric regions. The script is as follows:

```

1 clear;
2 clc;
3 a1=input("Enter the a1 value is:");
4 d=input("Enter the d value is:");
5 e=0.856*10^-12;
6 C=e*a1/d;
7 disp(C);
8 d = linspace(0, 10, 50);
9 y = C * exp(-0.1 * d);
10 plot2d3(d,y);
11 xlabel("d");
12 ylabel("capacitance");
13 title("Parallel plate capacity");
14 a2=input("Enter the a2 value is:");
15 d1=input("Enter the d1 value is:");
16 d2=input("Enter the d2 value is:");
17 d3=input("Enter the d3 value is:");
18 er1=input("Enter the er1 value is:");
19 er2=input("Enter the er2 value is:");
20 er3=input("Enter the er3 value is:");
21 C1=e*a2/((d1/er1)+(d2/er2)+(d3/er3));
22 disp(C1);

```



exp 6:-

$$a_1 = 3 \text{ cm}^2$$

$$a_2 = 4 \text{ cm}^2$$

$$d = 3 \text{ mm}$$

$$d_1 = 4 \text{ mm}$$

$$d_2 = 5 \text{ mm}$$

$$d_3 = 6 \text{ mm}$$

$$\epsilon r_1 = 1$$

$$\epsilon r_2 = 2$$

$$\epsilon r_3 = 3$$

(i) parallel plate capacitor

$$C = \frac{\epsilon A}{d}$$

$$C = \frac{(8.854 \times 10^{-12}) \times 3}{3}$$

$$C = 8.854 \times 10^{-12} \text{ F}$$

(ii) parallel plate capacitor with three dielectric layers

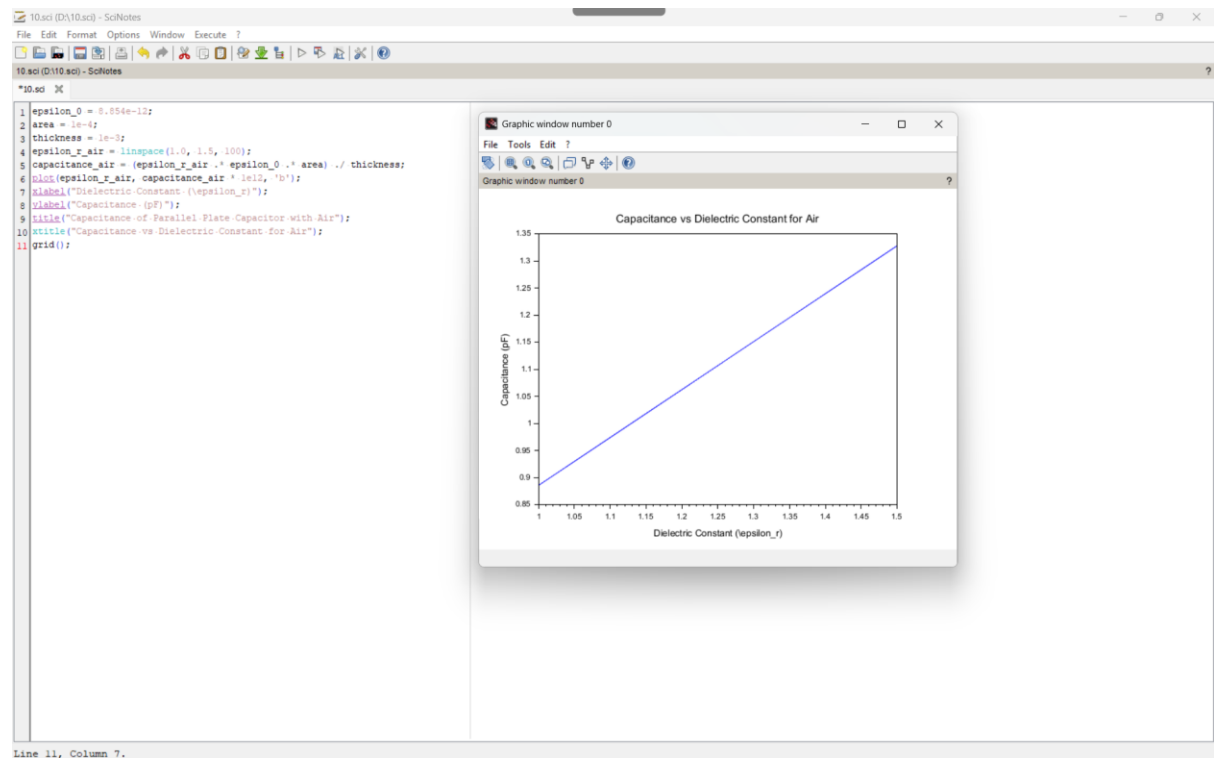
$$C = \frac{\epsilon A}{\left(\frac{d_1}{\epsilon r_1} + \frac{d_2}{\epsilon r_2} + \frac{d_3}{\epsilon r_3} \right)}$$

$$= \frac{35.416 \times 10^{-12}}{4/1 + 5/2 + 6/3}$$

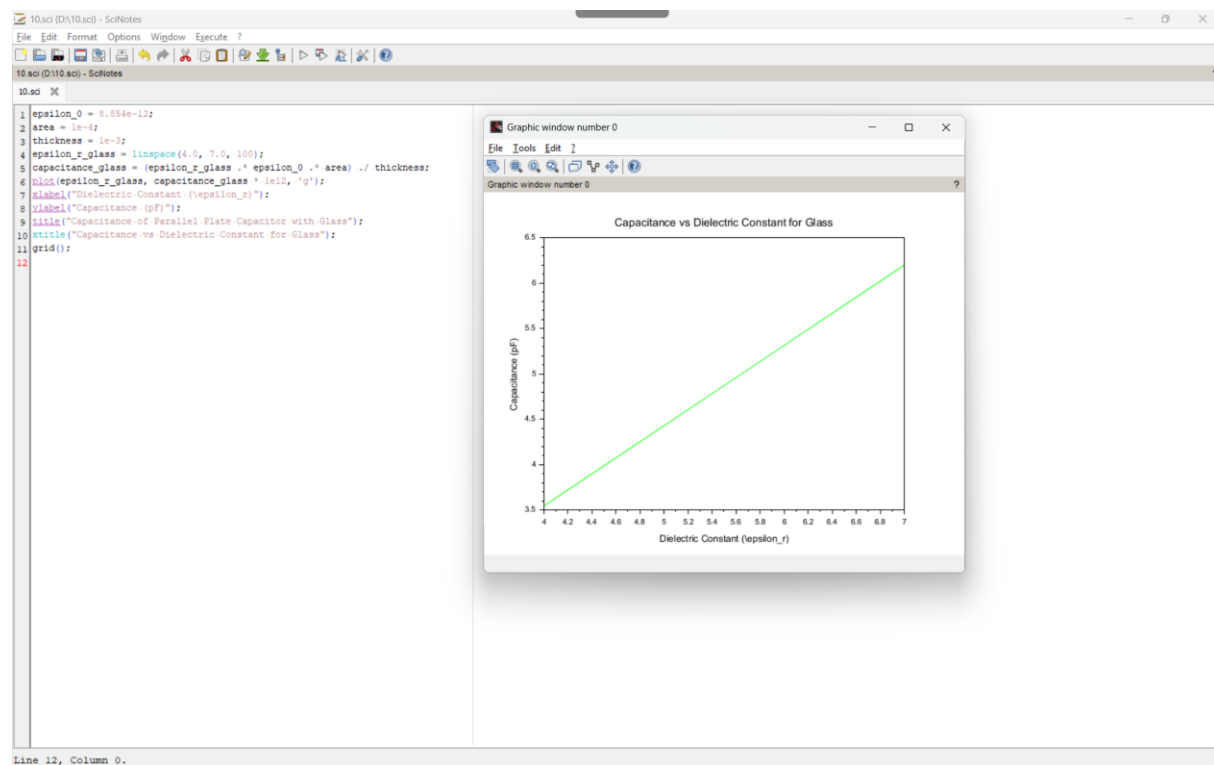
$$= \frac{35.416 \times 10^{-12}}{8.5}$$

$$C = 4.167 \times 10^{-12} \text{ F}$$

Case 1. AIR



Case 2. Glass:



Case 3. Teflon:

